

TANISH PAL

Software Developer — Full-Stack Developer — C++ — Python — Data Structures & Algorithms

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology (NIT Jalandhar)
B.Tech, Electronics and Communication Engineering

Nov 2022 – June 2026
CGPA: 7.21/10

EXPERIENCE

Basil Infotech

July – Dec 2025

Software Development Intern

Remote

- Developed the **Client Database** and **Lead Management** modules of a CRM/ERP system, covering client records, lead tracking, and centralized data storage.
- Implemented automated reporting workflows over client and lead data, replacing manual tracking with generated reports.

A2IT

June – July 2025

Software Development Intern

Remote

- Built **10+** **responsive React** UI components for the Cheffer app, improving cross-device compatibility; optimized state management to cut unnecessary re-renders.
- Integrated **REST APIs** with the backend team to connect the client and server data flow.

PROJECTS

Mock Mate 🔗 | *Next.js 14, Gemini API, Neon PostgreSQL, Drizzle ORM, Clerk, Web Speech API*

Live Demo 🔗

- Built an AI mock-interview platform using **Google Gemini** to generate role-specific questions and return per-answer scores, model answers, and written feedback.
- Added voice interviews with the browser **Web Speech API** (speech-to-text and text-to-speech), avoiding paid third-party audio services.
- Modeled interview sessions, question banks, and performance history in **Neon PostgreSQL** with **Drizzle ORM**; kept DB credentials server-side via **React Server Actions** and implemented **authentication and role-based route protection** with **Clerk**.

ChessAnalyzer 🔗 | *React, TypeScript, Stockfish 18 (WASM), Web Workers, Zustand, Vitest*

Live Demo 🔗

- Built a fully client-side chess analysis tool running **Stockfish 18 in WebAssembly**, with engine search offloaded to **Web Workers** and UCI output parsed asynchronously to keep the UI responsive.
- Implemented Chess.com-style accuracy scoring and a 10-tier move classification (Brilliant, Great, Missed Win, etc.) using a tanh-based win-probability model.
- Built an opening-book engine covering **110+** **variations** with prefix-set hash lookups for $O(1)$ ECO identification; wrote **61 unit tests** in Vitest for move classification and the UCI pipeline.

KnowYourProduct 🔗 | *React, Node.js, Express, MongoDB, JWT, Tailwind CSS*

Live Demo 🔗

- Built a full-stack food-scanning app: a browser **barcode scanner** (HTML5-QRCode) maps products to nutrition, allergen, and ingredient-concern data.
- Designed a **RESTful API** (Express, MongoDB) with Mongoose text indexes for product search and **JWT authentication and authorization** for personalized user dashboards.

Core Banking System 🔗 | *C++17, STL, File I/O, OOP, LLD*

Live Demo 🔗

- Built a PIN-authenticated banking CLI in C++17 (accounts, deposits, withdrawals, transfers) with a file-based persistence layer and append-only audit log that survives restarts; compiles warning-free under **-Wall -Wextra -Wpedantic**.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript (ES6+), TypeScript, SQL

Frameworks & Libraries: React.js, Next.js, Redux, Zustand, Node.js, Express.js, Tailwind CSS, Responsive Web Design

Backend & APIs: RESTful API Design, JWT Authentication, Role-Based Access Control, Server Actions, Postman

Testing & Automation: Unit Testing (Vitest), Pytest, Requests, Test Automation, API Testing, Logging, JSON

Databases: PostgreSQL, MySQL, MongoDB, Drizzle ORM, Mongoose

DevOps & Cloud: Docker, GitHub Actions (CI/CD), Linux, Nginx, AWS (EC2, S3), Cloudflare R2, Backblaze B2

Tools & APIs: Git, GitHub, Vercel, Render, VS Code, OpenAI API, Google Gemini API

Data Analysis & Visualization: Power BI, Power Query, DAX, Excel, Pandas, NumPy, Matplotlib, Tableau

Core CS: Data Structures & Algorithms, OOP, Low-Level Design, Operating Systems, DBMS

ACHIEVEMENTS

- **LeetCode:** peak rating 1726; solved 500+ DSA and Dynamic Programming problems (top percentile, Contest 479).
- **Smart India Hackathon** participant (Government of India); **GDSC NITJ Hackathon** finalist among 50+ teams.